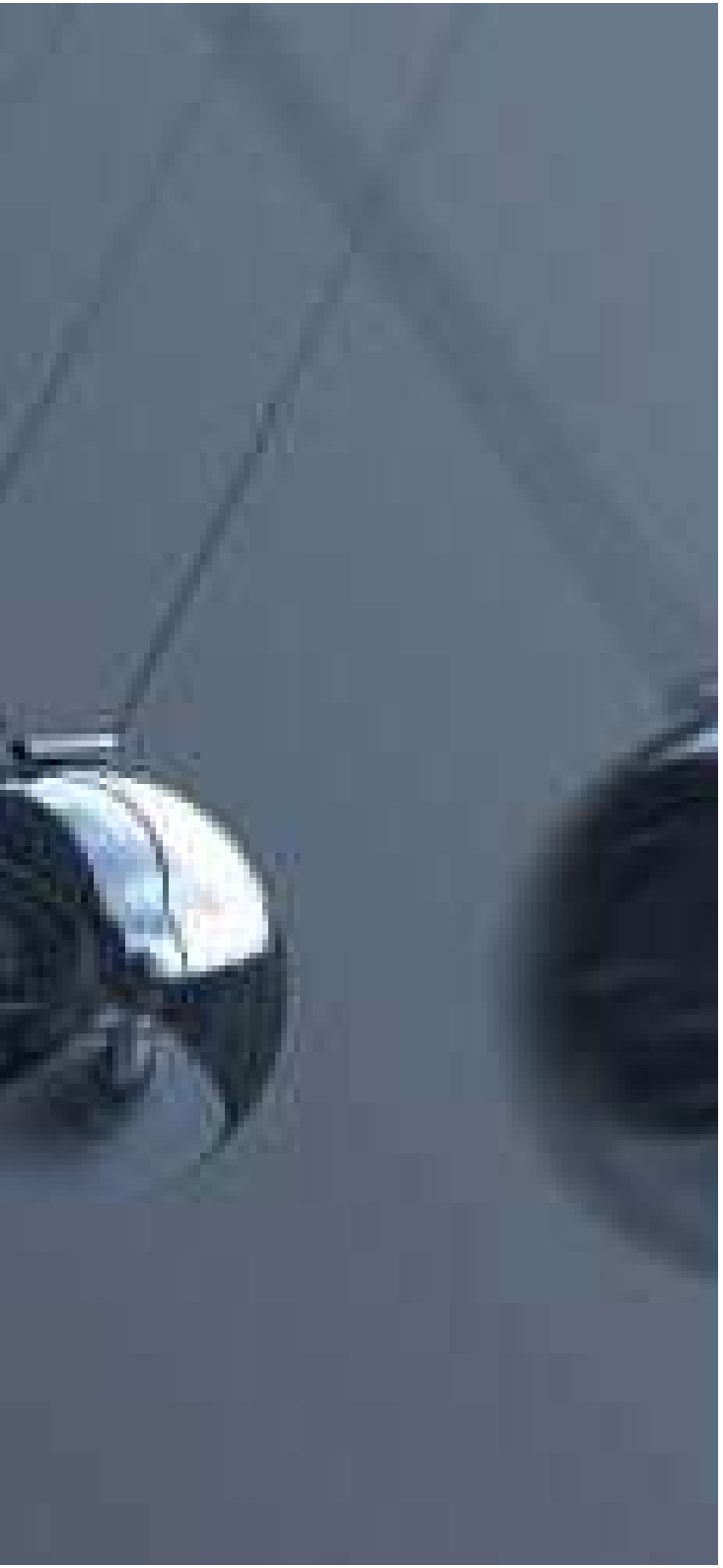
Digital Image Procesing



Grayscale to RGB Conversion

- convert an color image into a grayscale image. There are two methods to convert it.
  - Average method
  - Weighted method or luminosity method

Average method

- Grayscale = (R + G + B)/ 3





Weighted method or luminosity method

- grayscale = ((0.3 * R) + (0.59 * G) + (0.11 * B)).



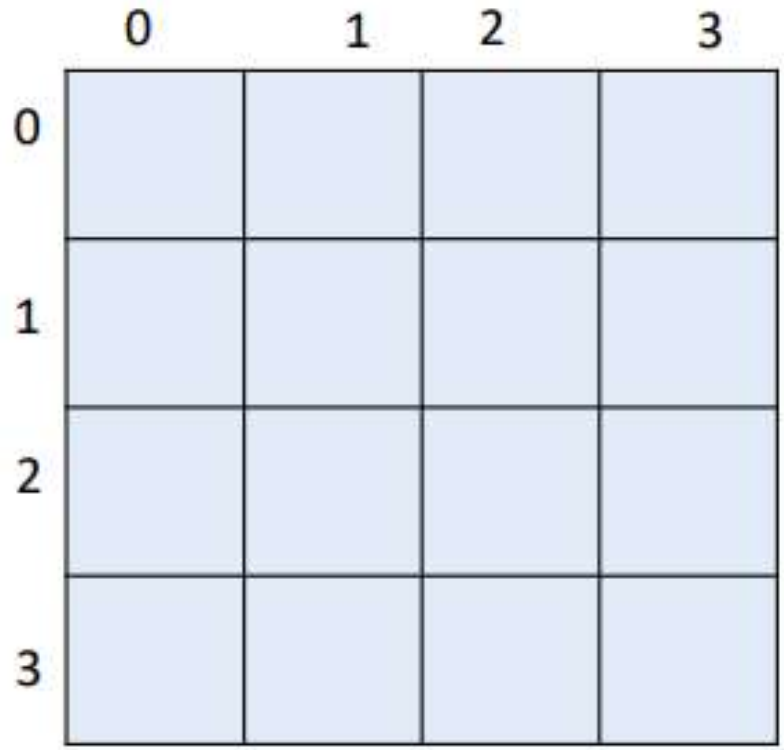


Resampling

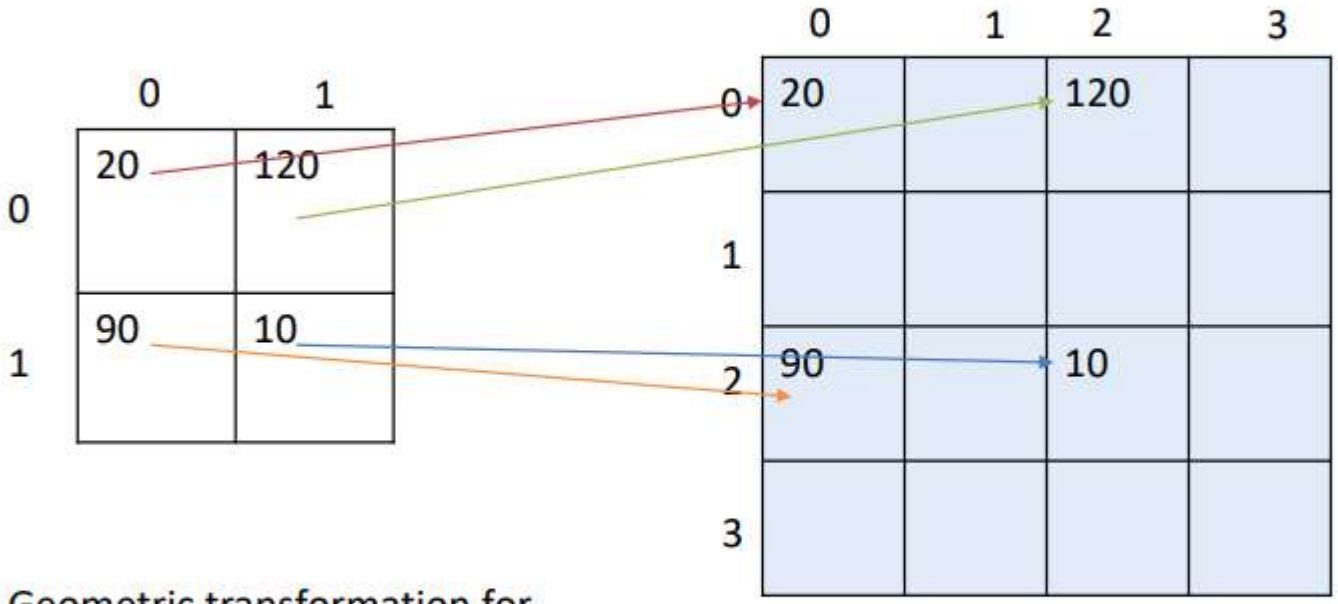
- Once the image is acquired.
- How to
  - Enlarge an image
  - Shrink an image
  - Zoom in
- Zooming Example:
  - Initial image size = 500 X 500
  - Required image size (= **X** 1.5) = 750 X 750

Geometric Transformation

	0	1
0	20	120
1	90	10



Geometric Transformation



Geometric transformation for mapping pixels.

(0,0) X 2 → (0,0)

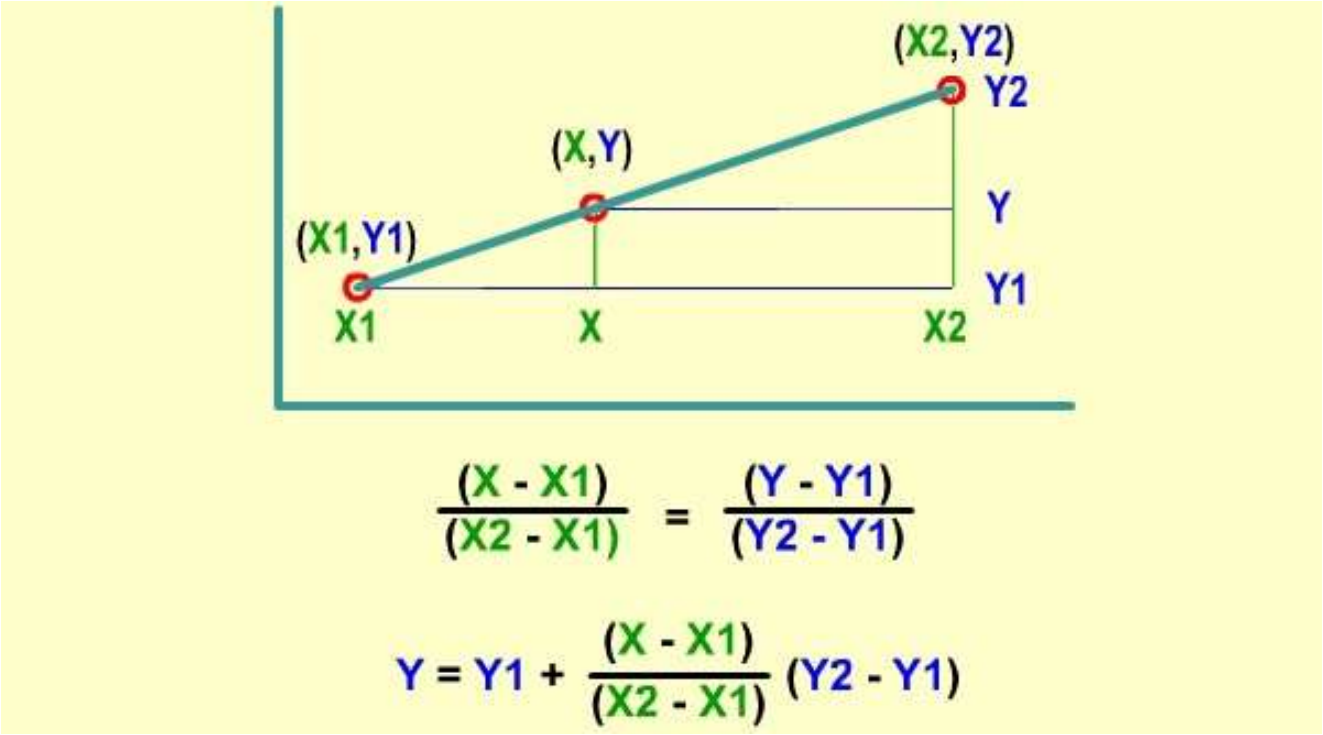
(0,1) X 2 → (0,2)

(1,0) X 2 → (2,0)

(1,1) X 2 → (2,2)

Image Interpolation

- Interpolation is the process of using known data to estimate values at unknown locations
- Ex. image of size 500 x 500 pixels has to be enlarged 1.5 times to 750 x 750 pixels



Geometric Transformation: Inverse lookup

	0	1
0	20	120
1	90	10

	0	1	2	3
0				
1				
2				
3				

Inverse mapping pixels.
(0,0) X 1/2 → (0,0)
(0,1) X 1/2 → (0,0.5)
(0,2) X 1/2 → (0,1)
(0,3) X 1/2 → (0,1.5)
…

1. Create an image of desired size
2. For each pixel in the new image calculate which pixel it corresponds to in the original image.
3. Use values from nearby pixel to guess missing values

Image Interpolation

- **Interpolation** — Process of using known data to estimate unknown values

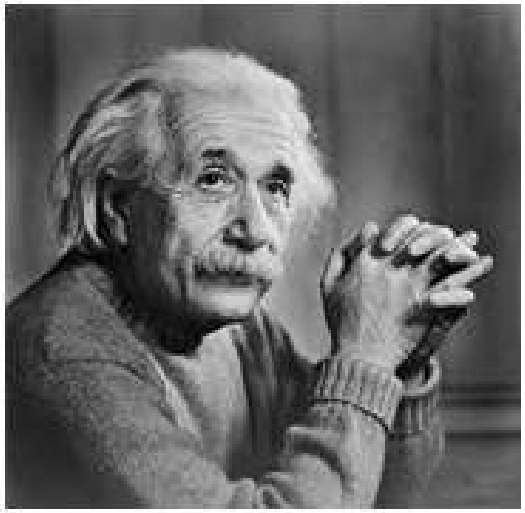
  e.g., zooming, shrinking, rotating, and geometric correction

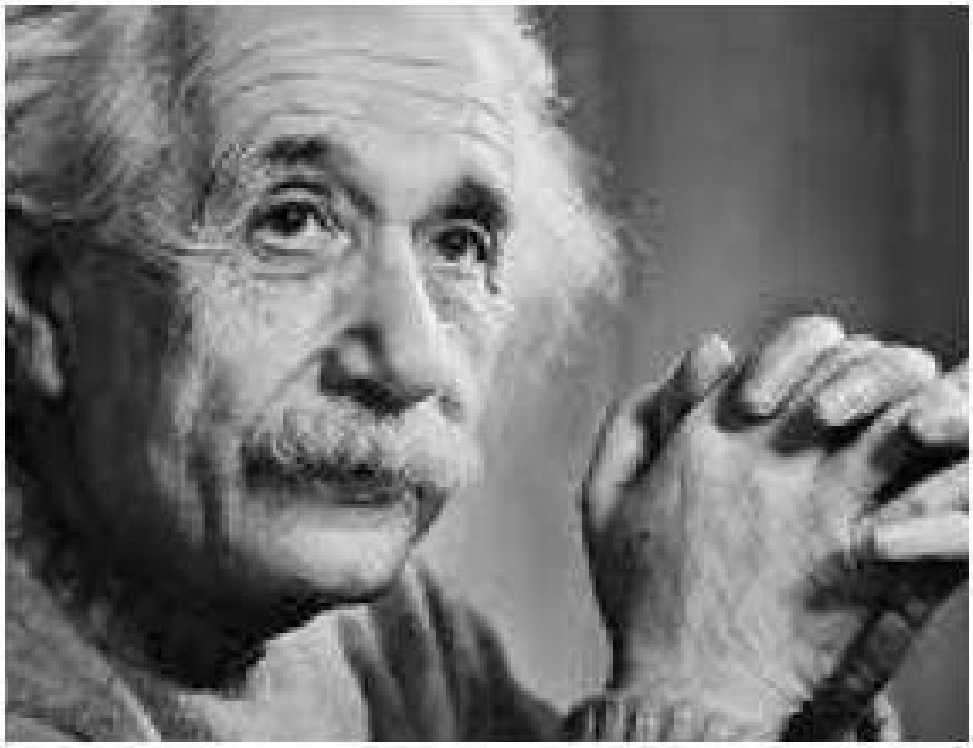
- **Interpolation** (sometimes called ***resampling***) — an imaging method to increase (or decrease) the number of pixels in a digital image.

  Some digital cameras use interpolation to produce a larger image than the sensor captured or to create digital zoom

Zooming

- Zooming simply means enlarging a picture in a sense that the details in the image became more visible and clear. Zooming an image has many wide applications ranging from zooming through a camera lens, to zoom an image on internet e.t.c.







a b c

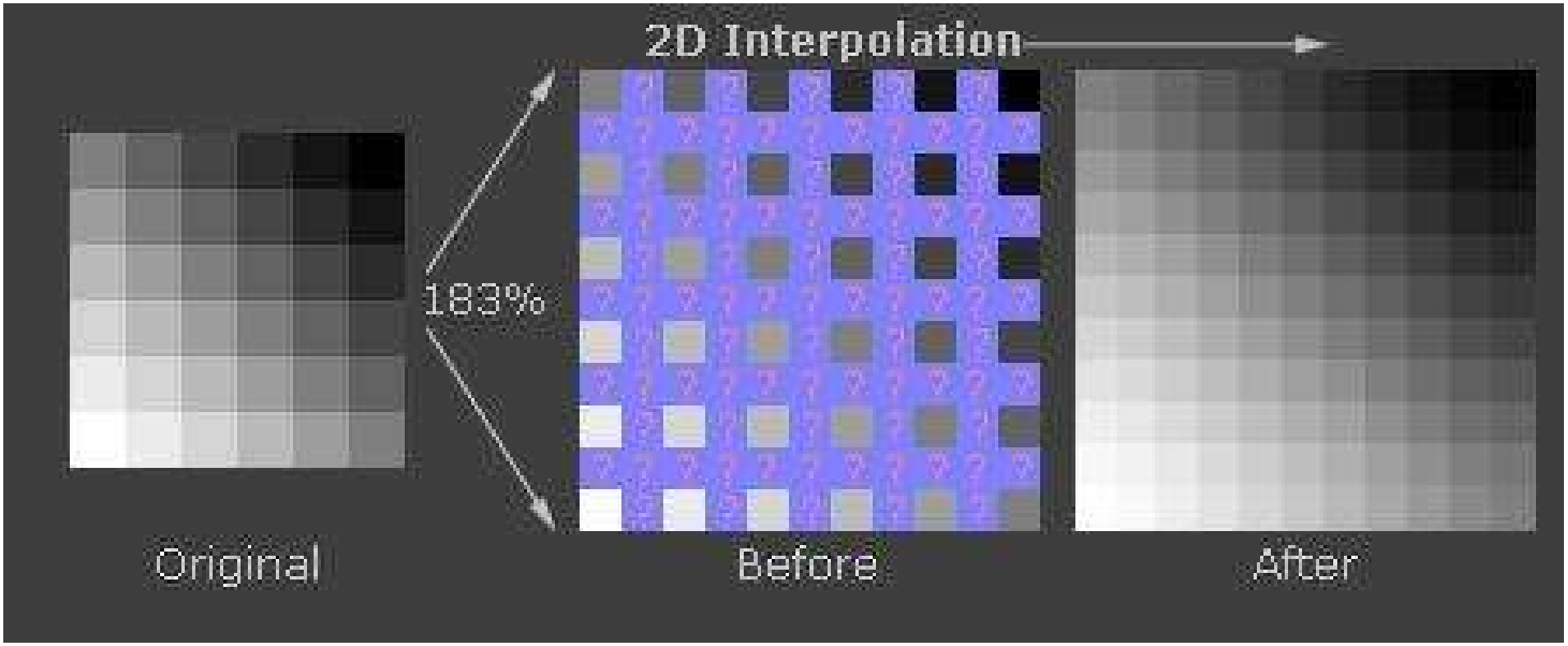
FIGURE 2.27

(a) Image reduced to 72 dpi and zoomed back to its original 930 dpi using nearest neighbor interpolation. This figure is the same as Fig. 2.23(d) . (b) Image reduced to 72 dpi and zoomed using bilinear interpolation. (c) Same as (b) but using bicubic interpolation.

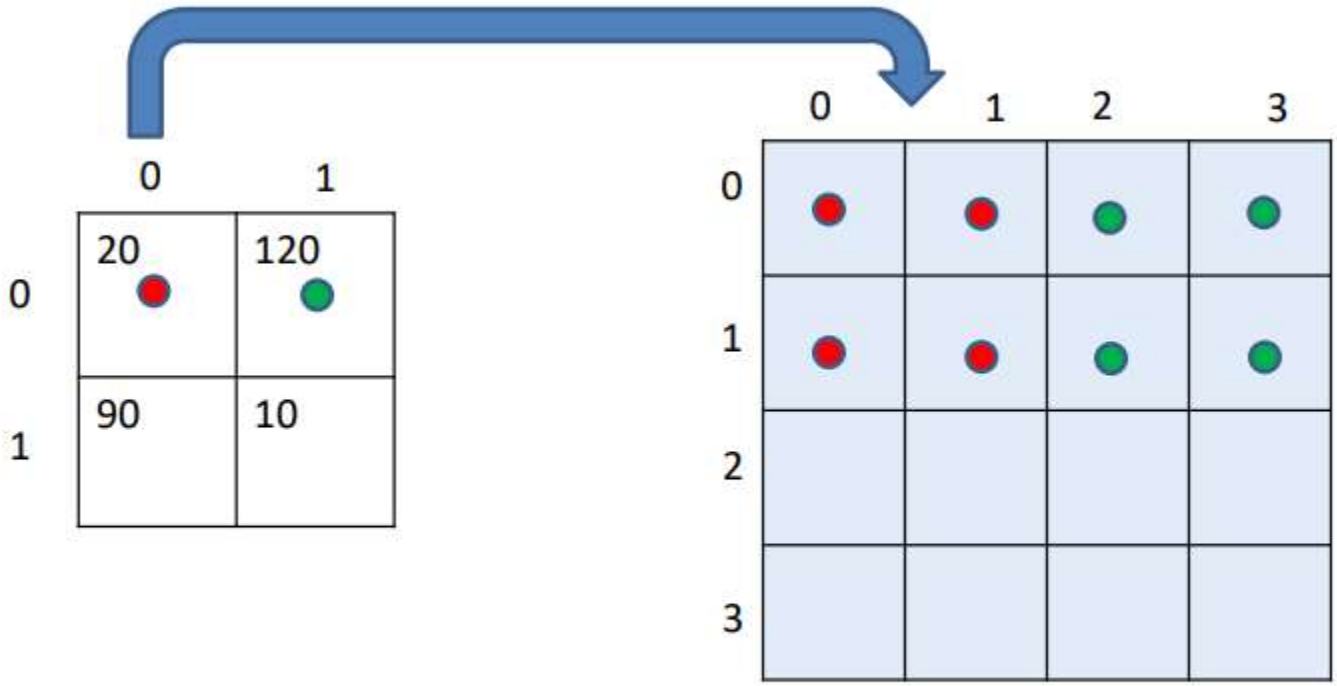
Zooming methods

- Although there are many methods that does this job, but we are going to discuss the most common of them here.
  - Pixel replication or (Nearest neighbor interpolation)
  - Zero order hold method
  - Zooming K times
  - Zooming By filter

Image Zooming



Nearest Neighbor

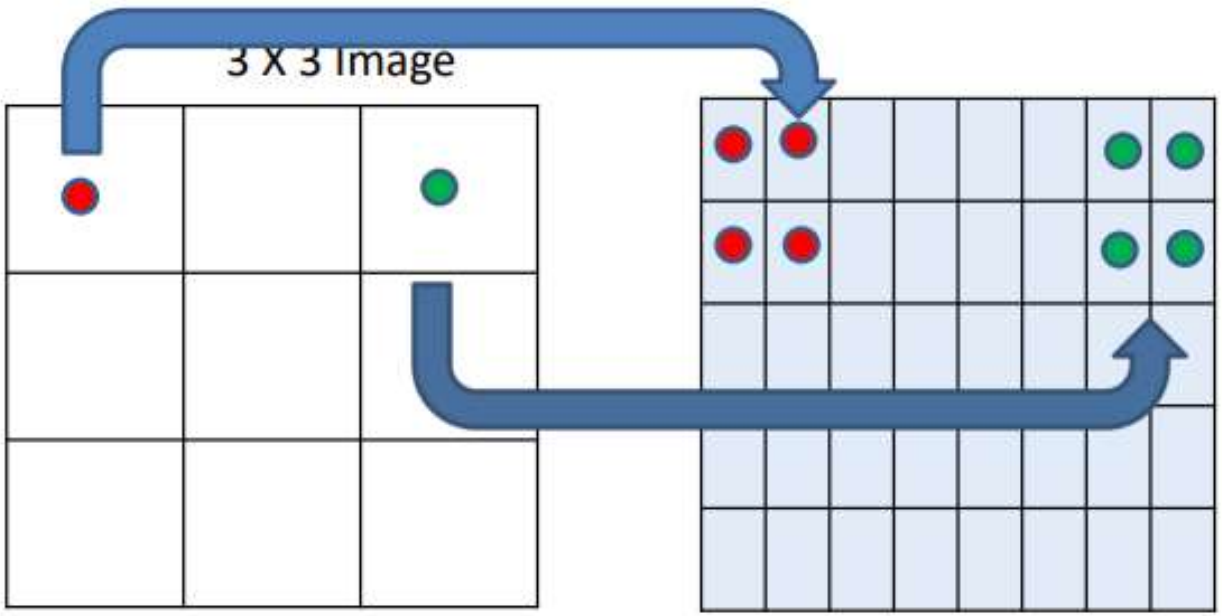


Inverse mapping pixels.
(0,0) X 1/2 → (0,0)
(0,1) X 1/2 → (0,0.5)
(0,2) X 1/2 → (0,1)
(0,3) X 1/2 → (0,1.5)

1. Create an image of desired size
2. For each pixel in the new image calculate which pixel it corresponds to in the original image.
3. Use **nearest pixel** values for missing values

Interpolation: Nearest Neighbor

5 X 8 Image



Fill in values preserving
spatial relationship

Zooming methods

Pixel replication or (Nearest neighbor interpolation)

Method 1: Pixel replication

- an image of 2 rows and 2 columns and want to zoom it twice or 2 times using pixel replication.

1	2
3	4

Row wise zooming:

1	1	2	2
3	3	4	4

Zooming methods

Pixel replication or (Nearest neighbor interpolation)

1	2
3	4

Row wise zooming:

1	1	2	2
1	1	2	2
3	3	4	4
3	3	4	4

Zooming methods

Pixel replication or (Nearest neighbor interpolation)

New image size:

- (Original image rows * zooming factor, Original Image cols * zooming factor)
- advantage : very simple
- The disadvantage output is very blurry

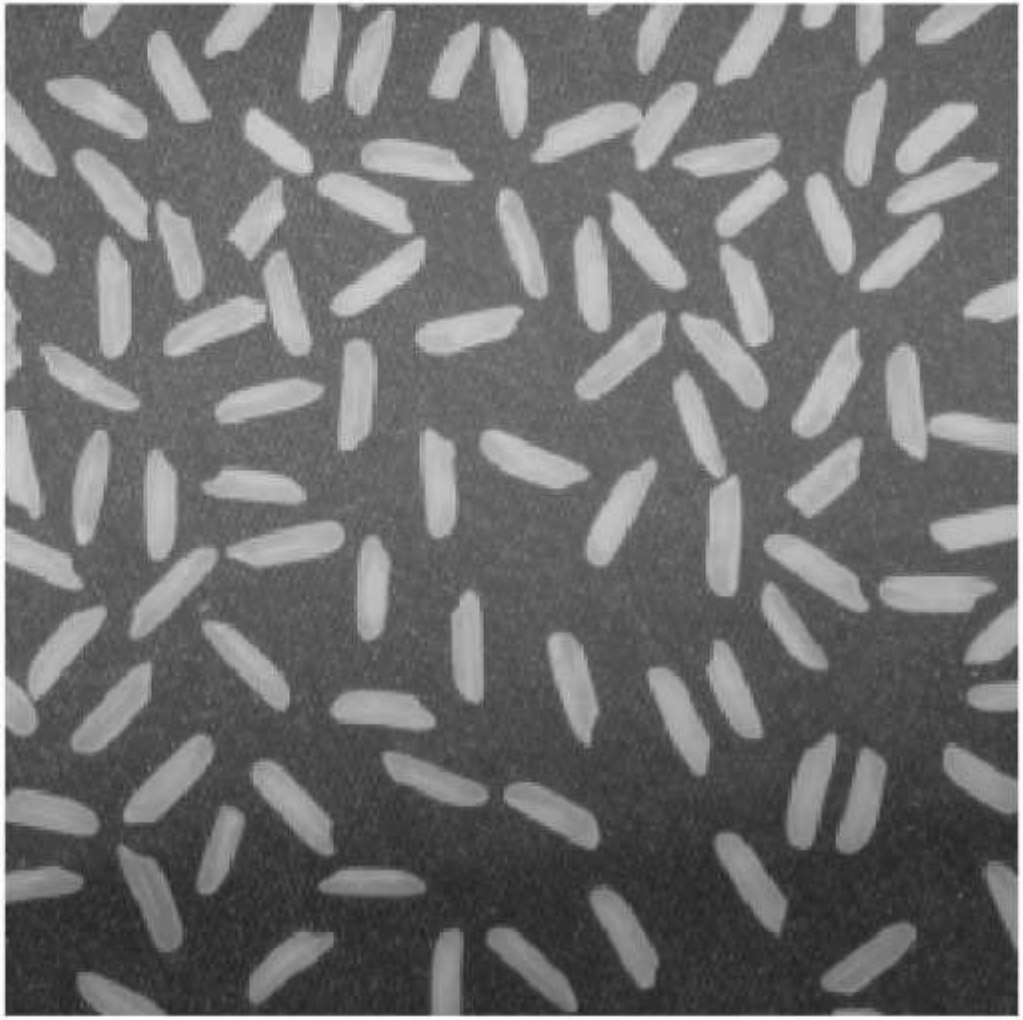
Interpolation: Nearest neighbor

original image



Original

nearest



Zooming methods

Method 2: Zero order hold

- It is also known as zoom twice. Because it can only zoom twice.
- Working:
- we pick two adjacent elements from the rows respectively and then we add them and divide the result by two, and place their result in between those two elements. We first do this row wise and then we do this column wise.

Zooming methods

Method 2: Zero order hold

An image 2 x 2:

1	2
3	4

Row wise zooming:

1	1	2
3	3	4

Zooming methods

Method 2: Zero order hold

Row wise zooming :

1	1	2
3	3	4

Column wise zooming :

1	1	2
2	2	3
3	3	4

Zooming methods

Method 2: Zero order hold

New image size

- As you can see that the dimensions of the new image are 3 x 3 where the original image dimensions are 2 x 2. So it means that the dimensions of the new image are based on the following formula

  (2(number of rows) minus 1) X (2(number of columns) minus 1)

Advantages and disadvantage.

- One of the advantage : does not create as blurry picture But it also has a disadvantage that it can only run on the power of 2 ; 2,4,8,16,32 and so on

Zooming methods

Method : K-Times zooming

- First of all, you have to take two adjacent pixels as you did in the zooming twice. Then you have to subtract the smaller from the greater one. We call this output (OP).
- Divide the output(OP) with the zooming factor(K). Now you have to add the result to the smaller value and put the result in between those two values.
- Add the value OP again to the value you just put and place it again next to the previous putted value. You have to do it till you place k-1 values in it.
- Repeat the same step for all the rows and the columns , and you get a zoomed images.

Zooming methods
Method : K-Times zooming

- **For example:**

  Suppose you have an image of 2 rows and 3 columns, which is given below. And you have to zoom it twice or three times.
- K in this case is 3. K = 3.
- The number of values that should be inserted is k-1 = 3-1 = 2.

15	30	15
30	15	30

Zooming methods
Method : K-Times zooming

Row wise zooming

- Take the first two adjacent pixels. Which are 15 and 30.
- Subtract 15 from 30. 30-15 = 15.
- Divide 15 by k. 15/k = 15/3 = 5. We call it OP
- Add OP to lower number. 15 + OP = 15 + 5 = 20.
- Add OP to 20 again. 20 + OP = 20 + 5 = 25.
- We do that 2 times because we have to insert k-1 values.
- Now repeat this step for the next two adjacent pixels

15	20	25	30	20	25	15
30	20	25	15	20	25	30

Zooming methods
Method : K-Times zooming

- After inserting the values, you have to sort the inserted values in ascending order, so there remains a symmetry between them.

15	20	25	30	20	25	15
30	20	25	15	20	25	30

15	20	25	30	25	20	15
30	25	20	15	20	25	30

Zooming methods
Method : K-Times zooming

- **Column wise zooming**
- The same procedure has to be performed column wise

Here what you got after all that.

15	20	25	30	25	20	15
20	21	21	25	21	21	20
25	22	22	20	22	22	25
30	25	20	15	20	25	30

Zooming methods
Method : K-Times zooming

- New image size, The formula thus is:

  (K (number of rows minus 1) + 1) X (K (number of cols minus 1) + 1)

  The dimensions of the original image were 2 X 3. And the dimensions of the new image are 4 x 7.

- **Advantages and disadvantages**

  advantage : improved result (less blurry)

  The only difficulty this algorithm has that it has to be sort in the end, which is an additional step, and thus increases the cost of computation.

Zooming by Filtering (Convolution):

This process requires a mathematical process to enlarge an image. This method required two steps:

1. Extend the image by adding rows and columns of zeros between the existing rows and columns.
2. Perform the convolution.
   image is extended as follows:

First we expand the image x ($N \times M$) through adding zero rows and zero columns:

$$x = \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ p & q & r & s \\ t & u & v & w \end{bmatrix} \rightarrow \quad y = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & a & 0 & b & 0 & c & 0 & d & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & e & 0 & f & 0 & g & 0 & h & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & p & 0 & q & 0 & r & 0 & s & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & t & 0 & u & 0 & v & 0 & w & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Original Image Array

$$\begin{pmatrix} 3 & 5 & 7 \\ 2 & 7 & 6 \\ 3 & 4 & 9 \end{pmatrix}$$

Image extended with zeros

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 5 & 0 & 7 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 7 & 0 & 6 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 4 & 0 & 9 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Convolution mask for first –order hold

$$\begin{pmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{pmatrix}$$

B) Shrinking

Shrinking may be viewed as undersampling. Image shrinking is performed by **row-column deletion**. For example, to shrink an image by one-half, we delete every other row and column.

$$\begin{bmatrix} 69 & 69 & 50 & 50 & 80 & 80 \\ 69 & 69 & 50 & 50 & 80 & 80 \\ 45 & 45 & 60 & 60 & 66 & 66 \\ 45 & 45 & 60 & 60 & 66 & 66 \\ 30 & 30 & 55 & 55 & 80 & 80 \\ 30 & 30 & 55 & 55 & 80 & 80 \end{bmatrix} = \begin{bmatrix} 69 & 69 & 50 & 50 & 80 & 80 \\ 45 & 45 & 60 & 60 & 66 & 66 \\ 30 & 30 & 55 & 55 & 80 & 80 \end{bmatrix} = \begin{bmatrix} 69 & 50 & 80 \\ 45 & 60 & 66 \\ 30 & 55 & 80 \end{bmatrix}$$

Original image | image with rows deleted | image with rows and columns deleted